

Printer Dover

Spruce version 11.2 -- spooler version 11.2

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Creation date: 8-Oct-81 15:24:59

For: Spruce

25 total sheets = 24 pages, 1 copy.

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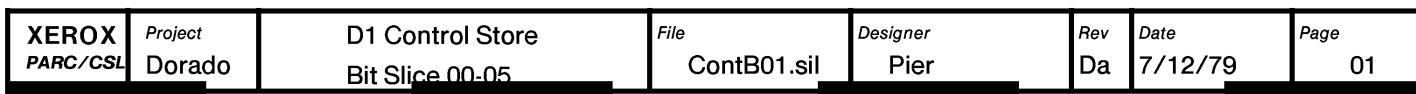
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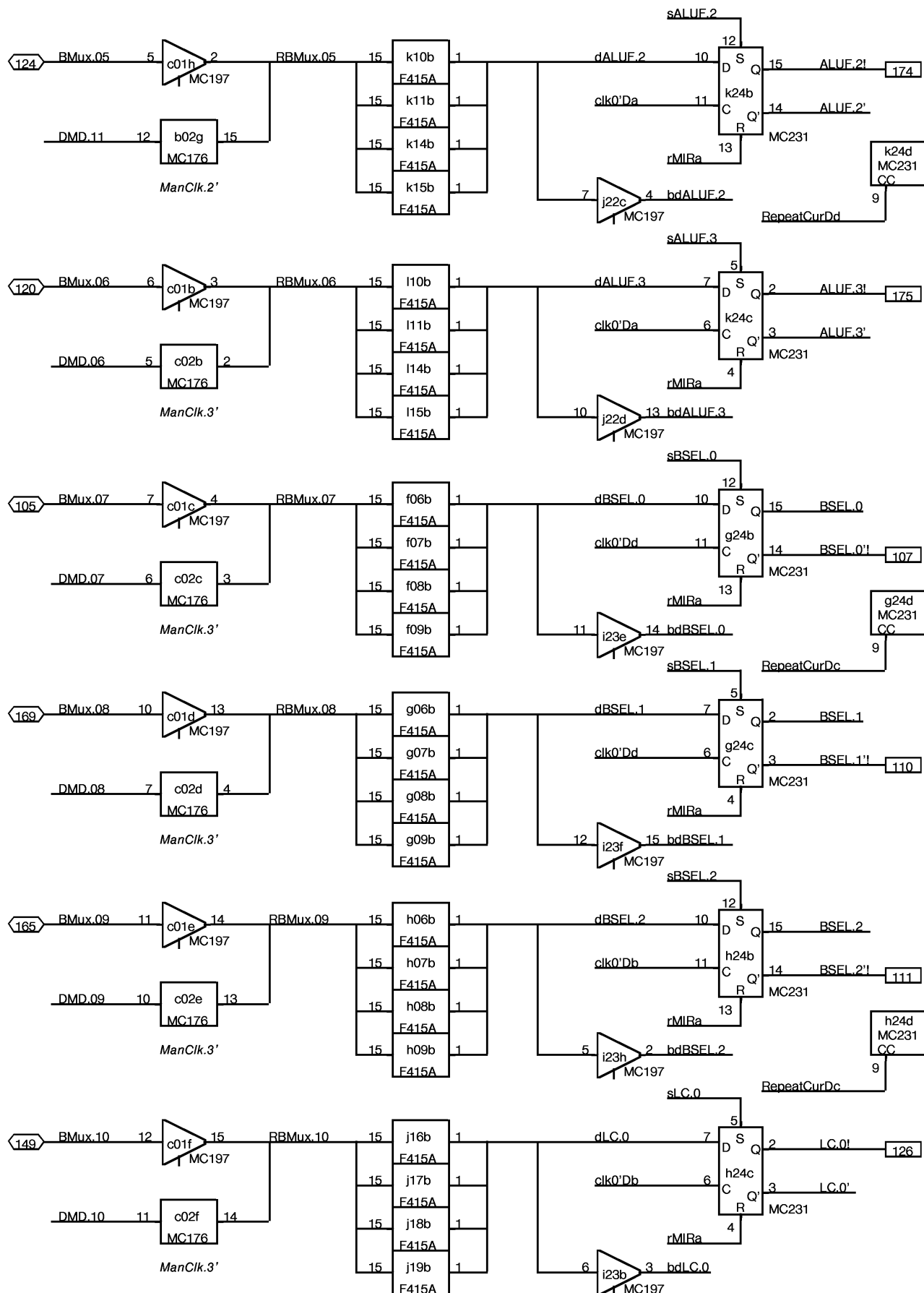
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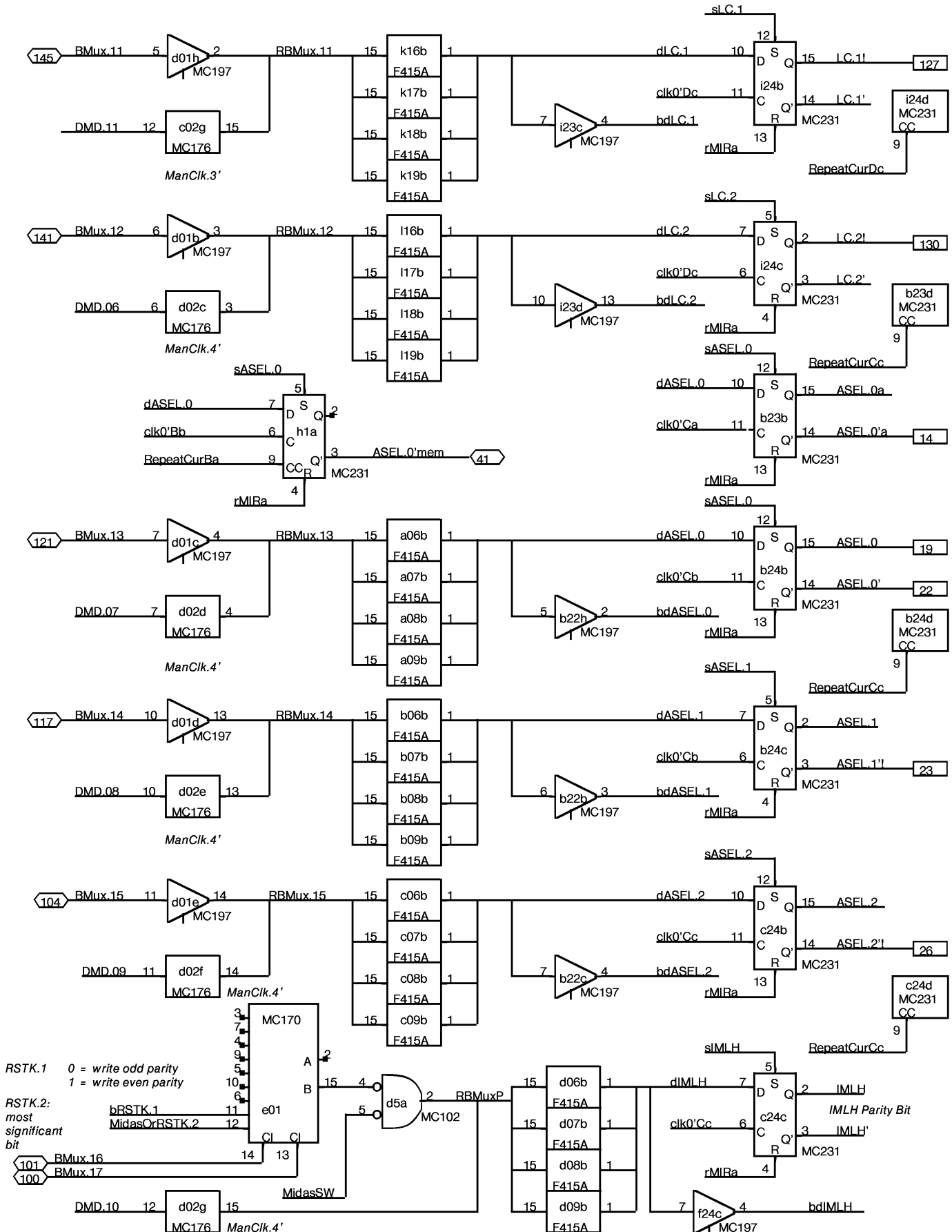
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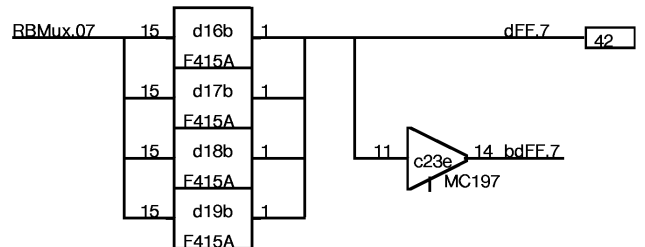
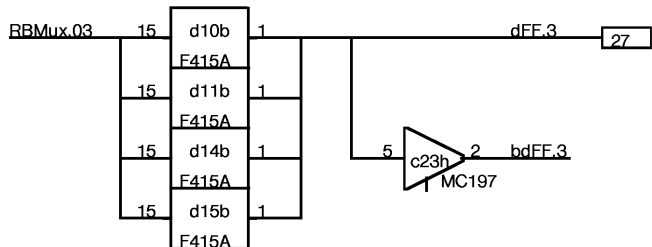
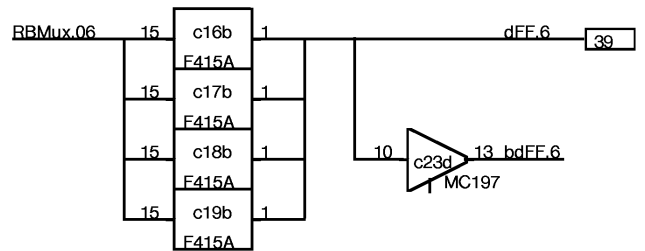
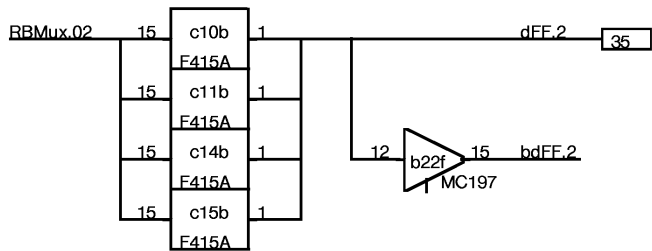
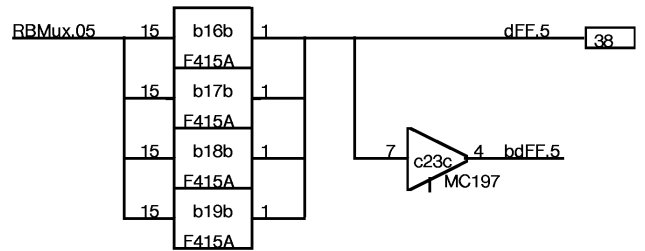
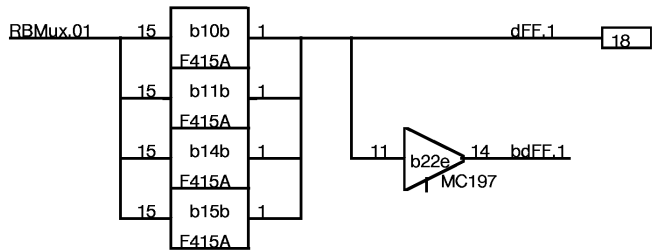
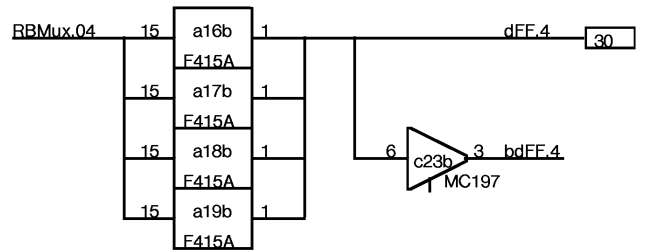
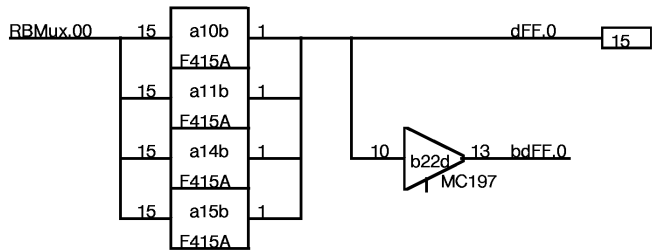
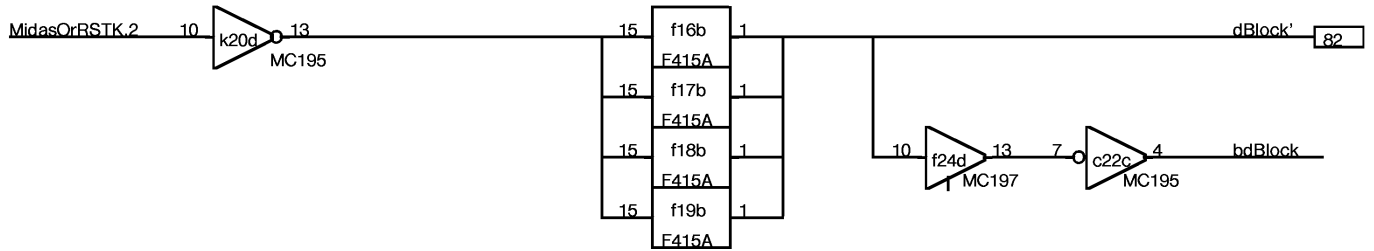
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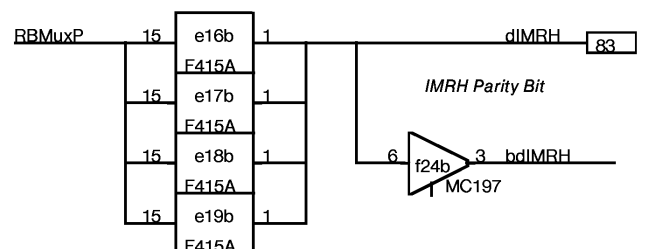
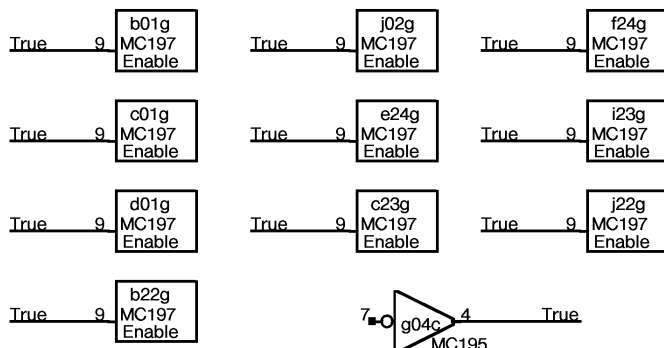
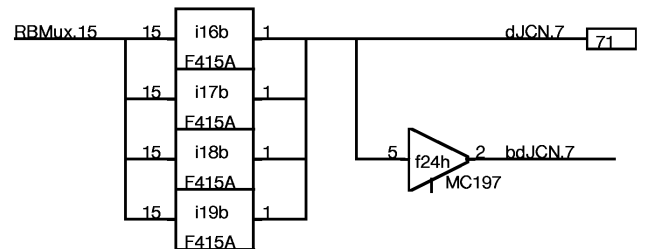
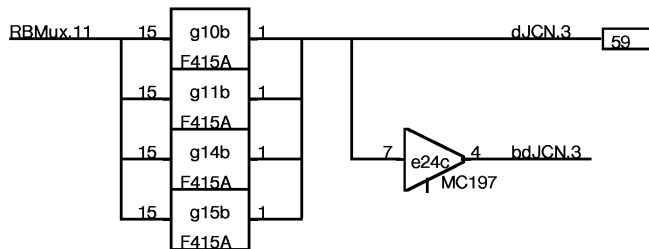
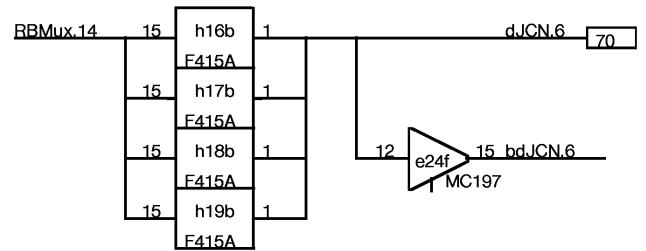
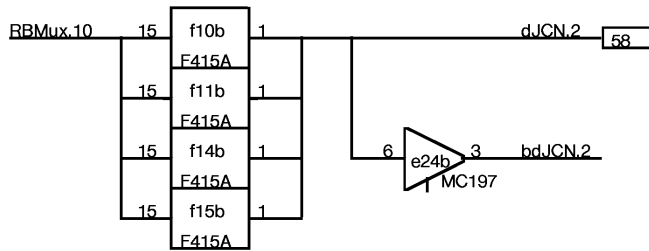
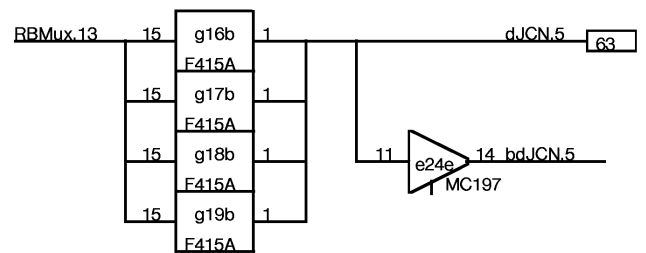
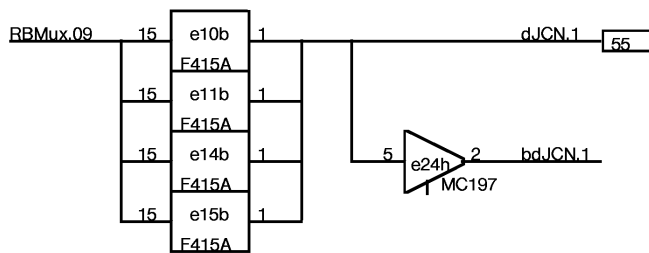
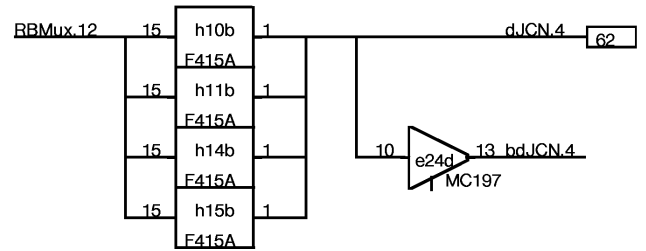
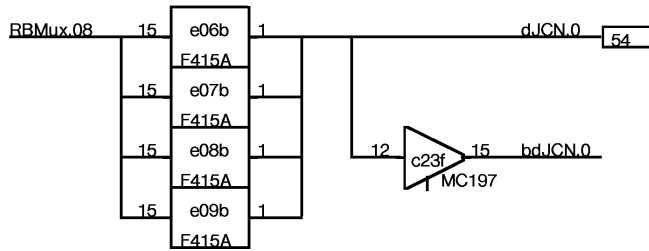
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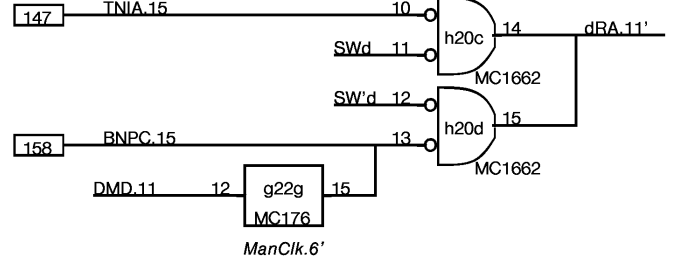
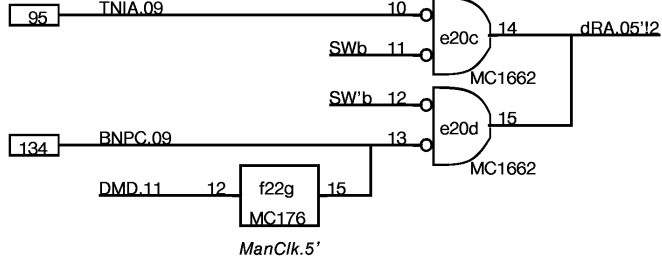
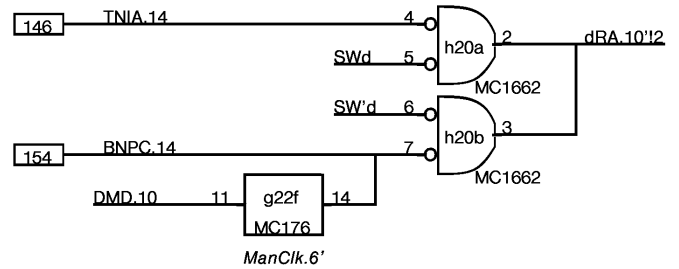
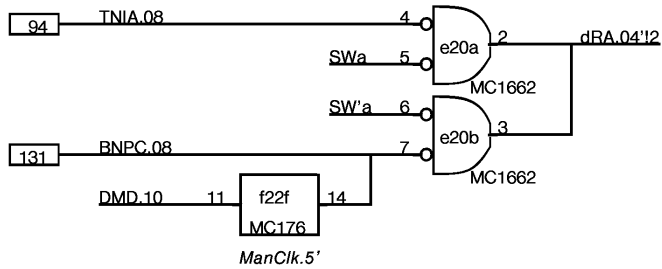
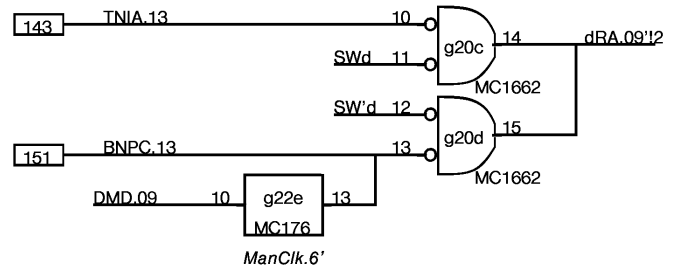
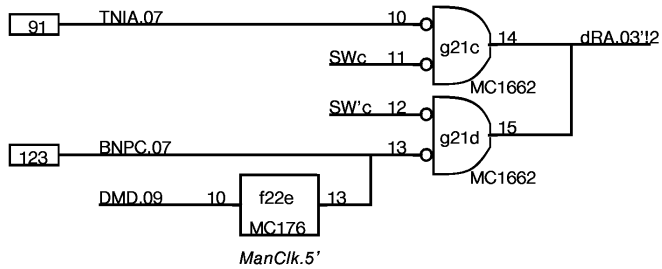
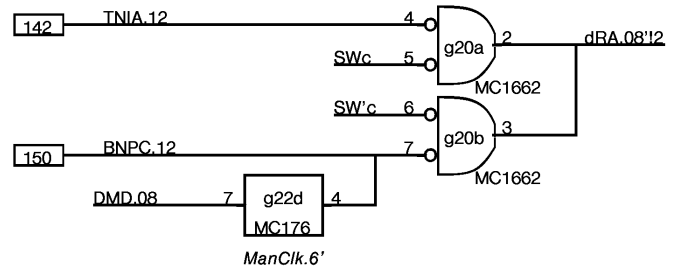
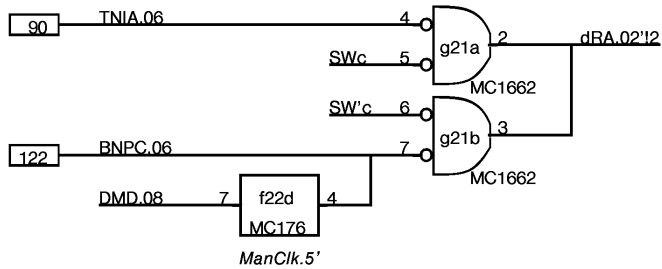
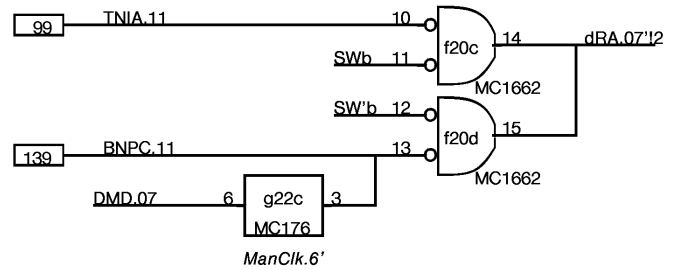
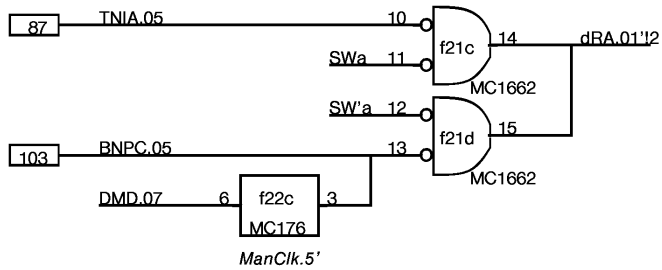
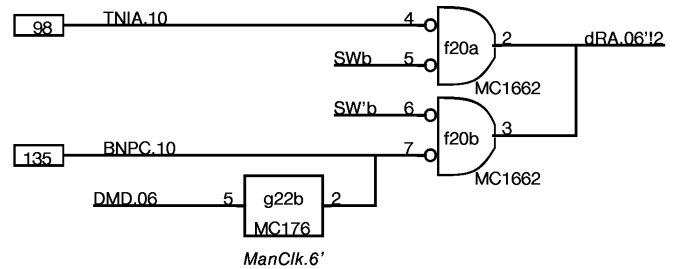
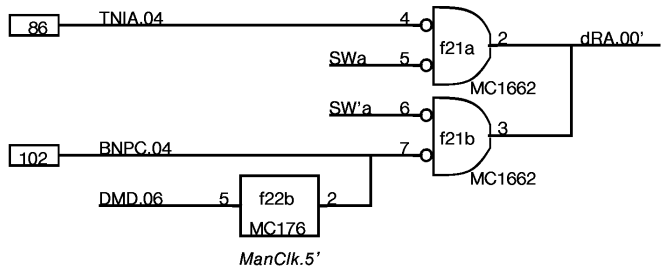


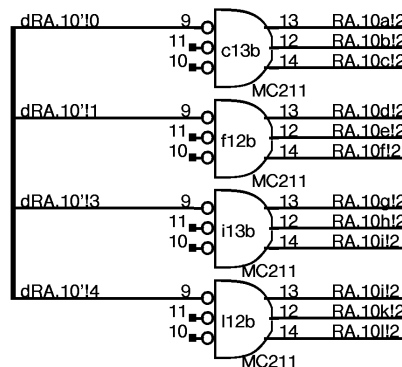
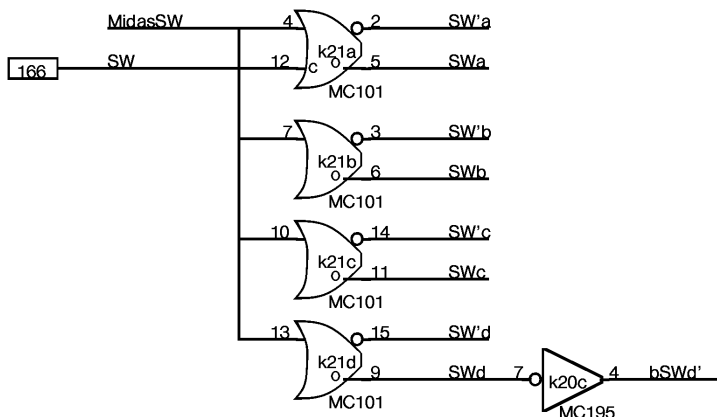
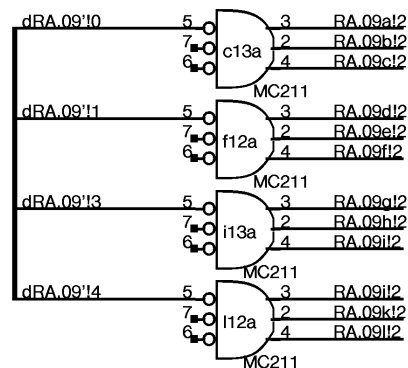
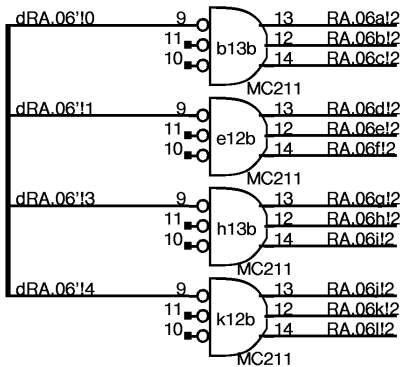
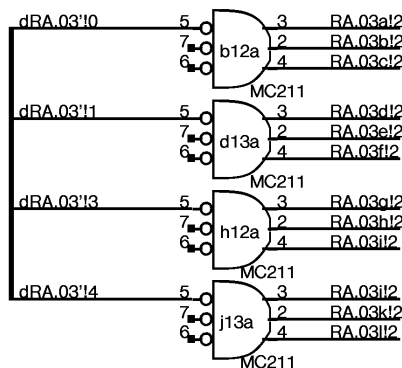
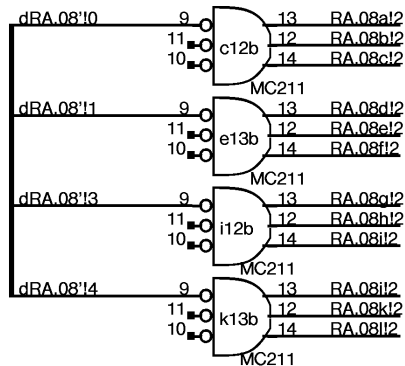
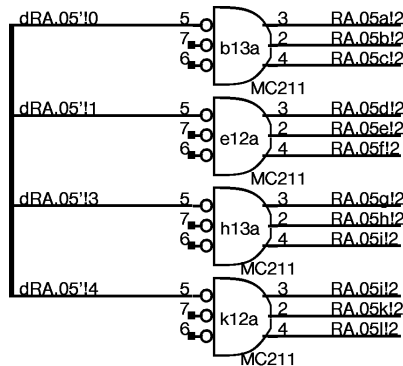
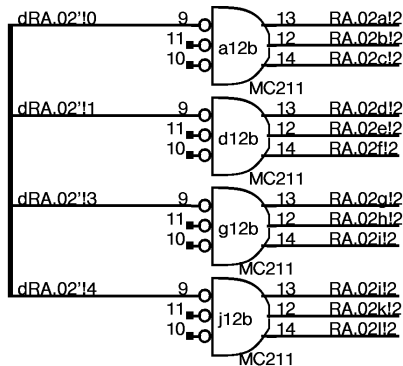
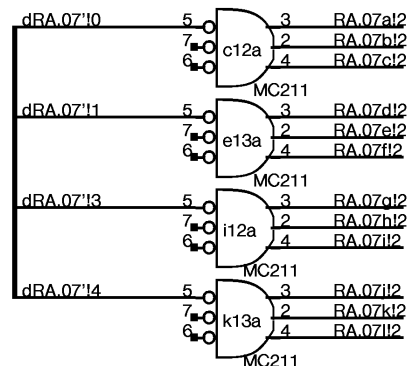
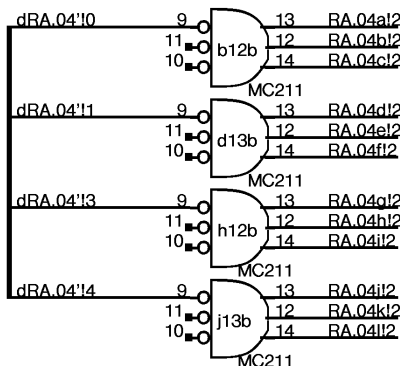
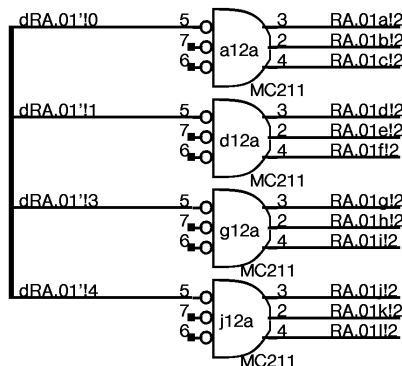


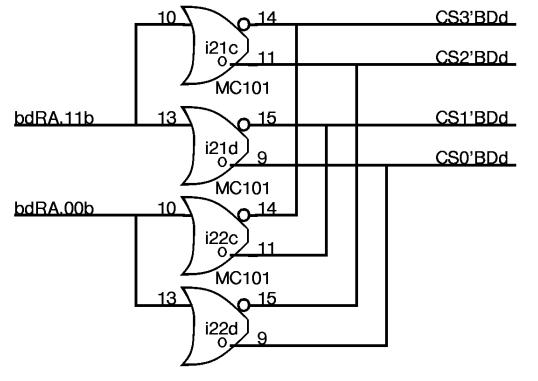
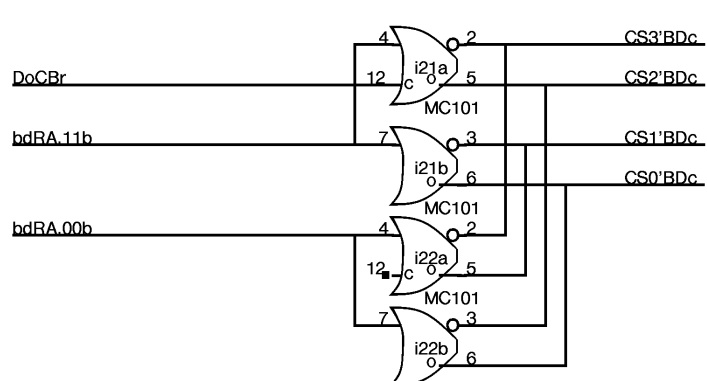
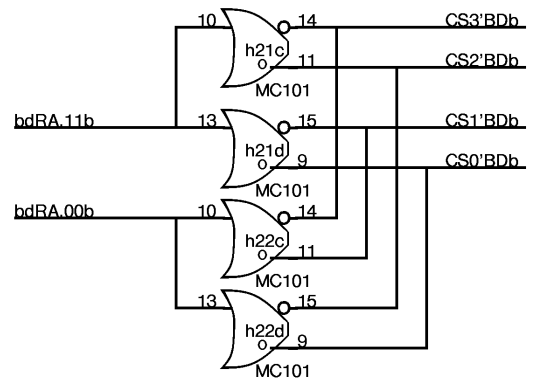
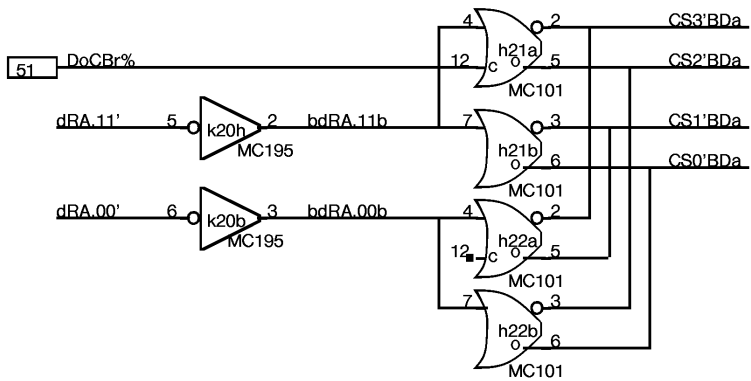
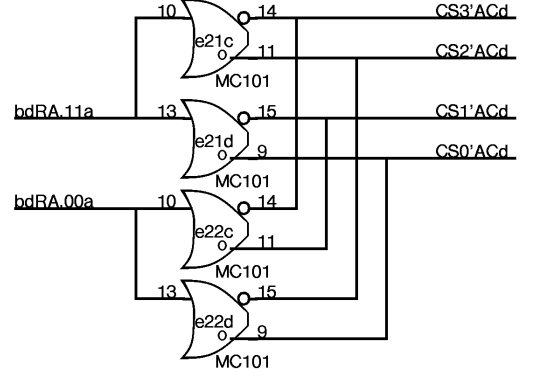
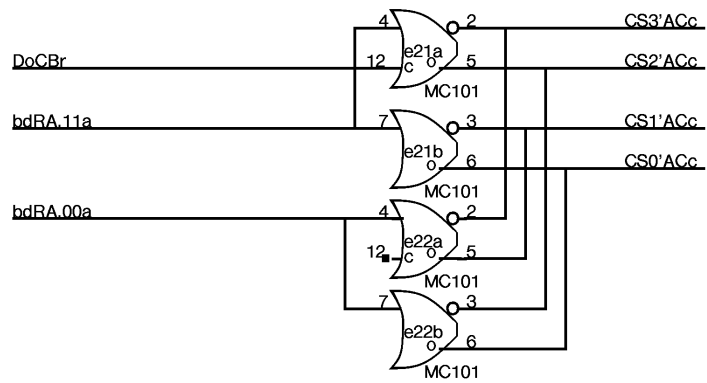
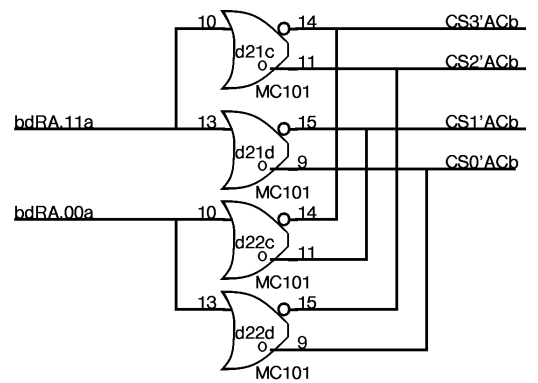
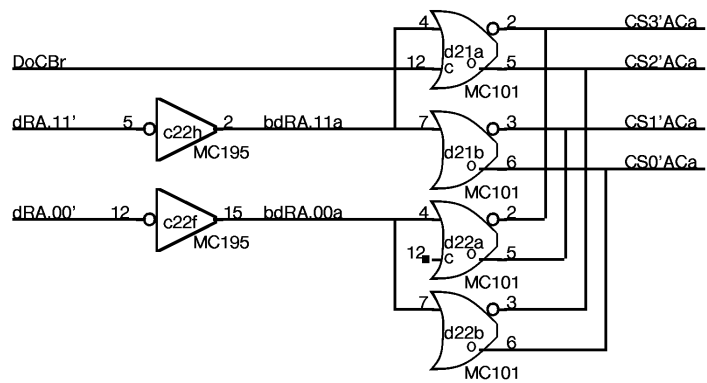


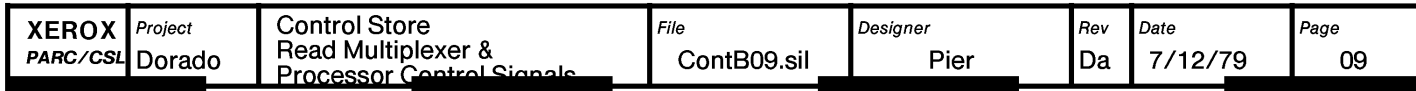
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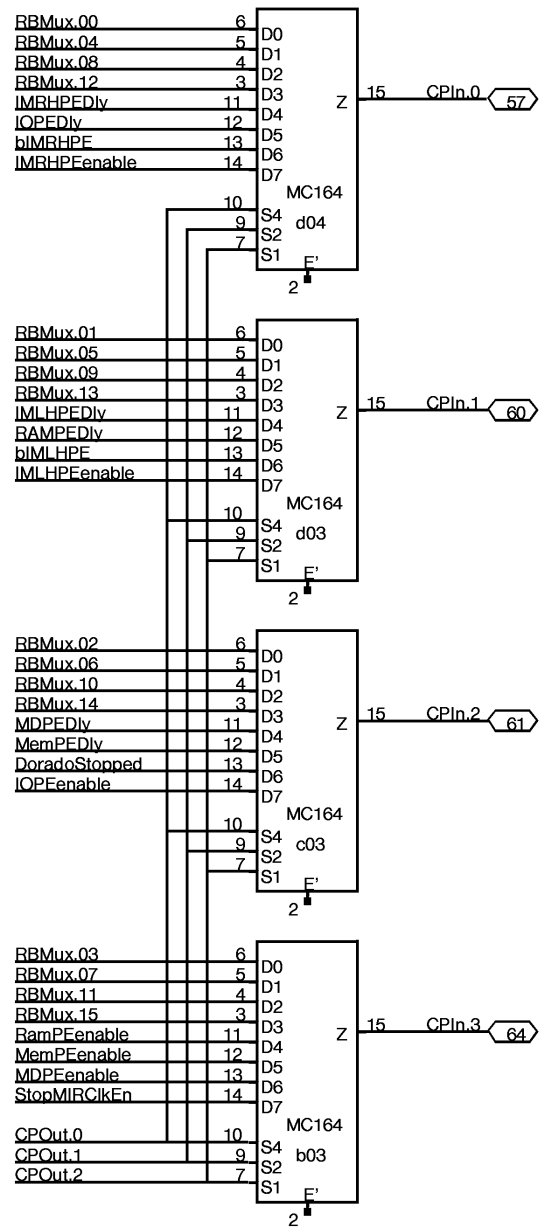
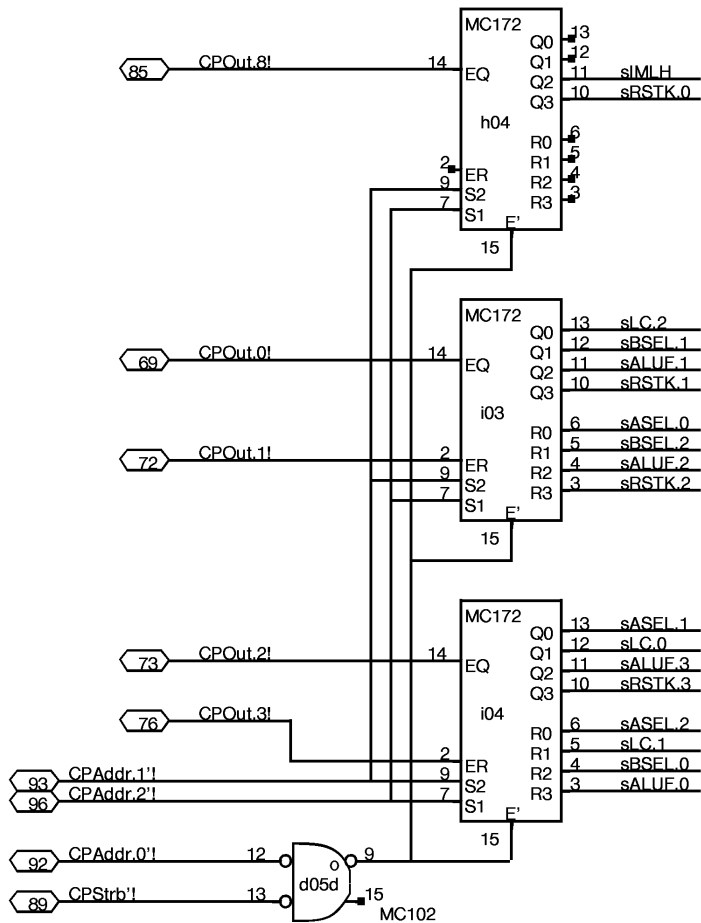
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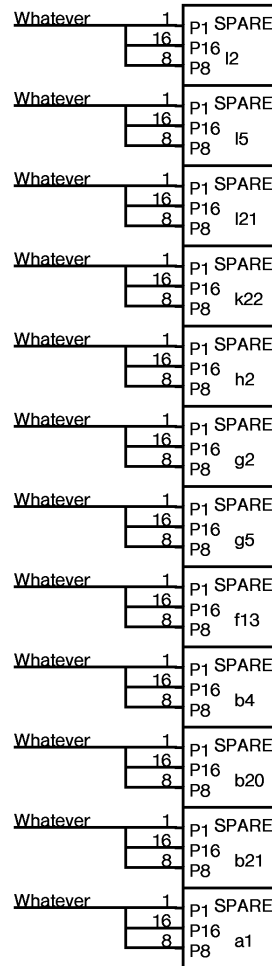
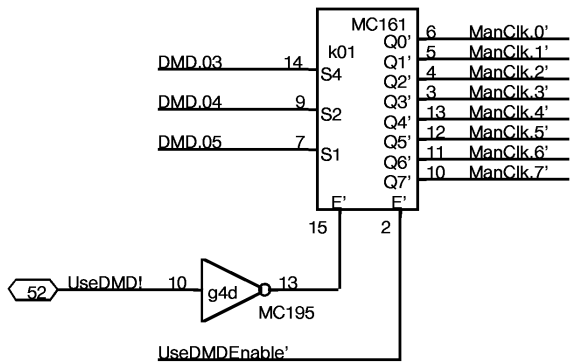
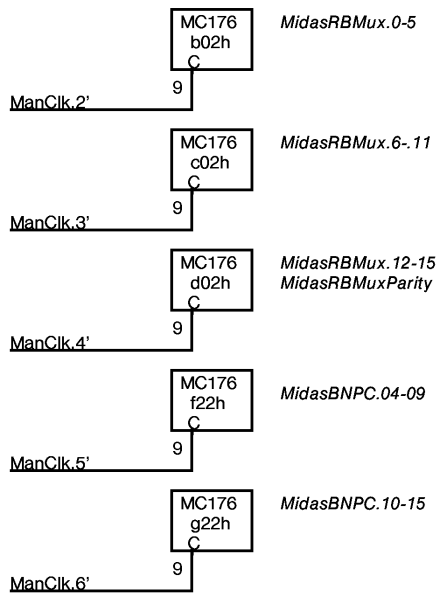
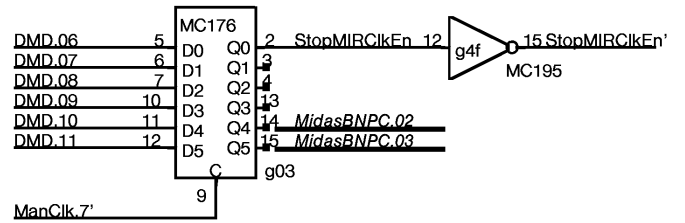
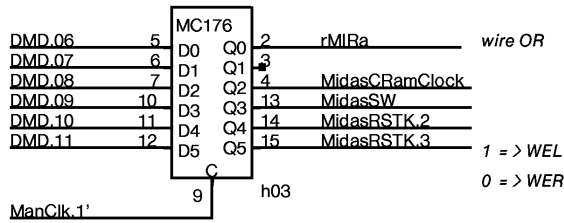
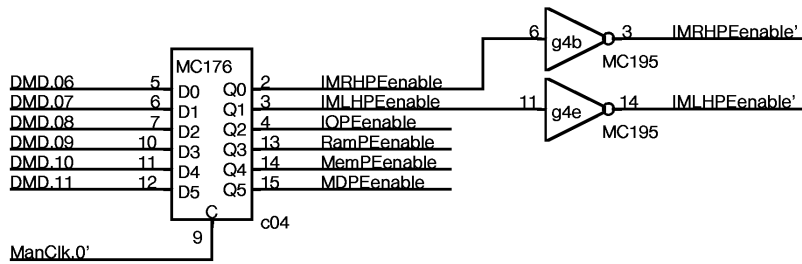




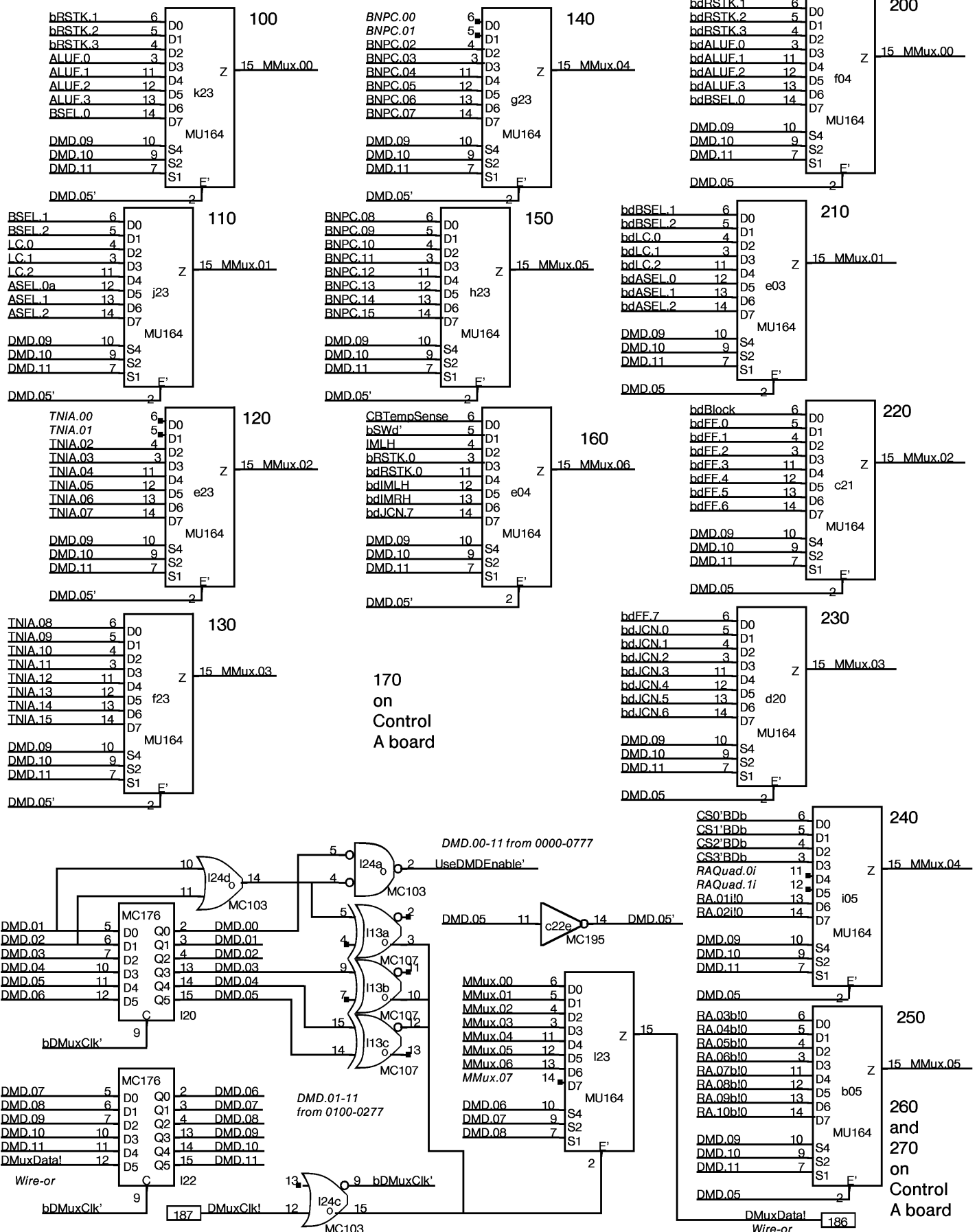








Multiwire
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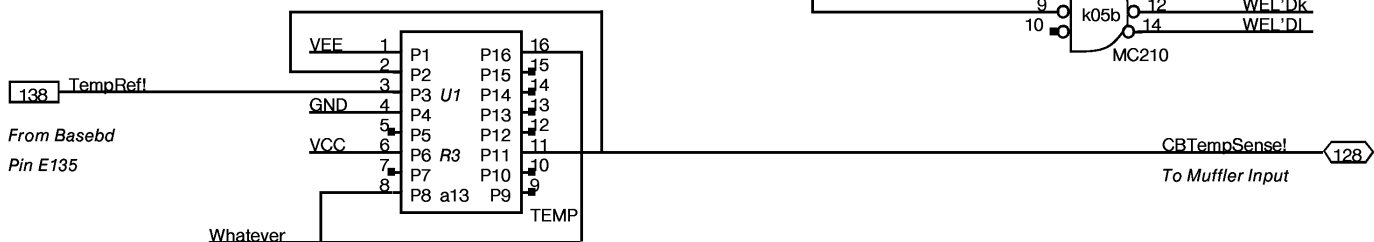
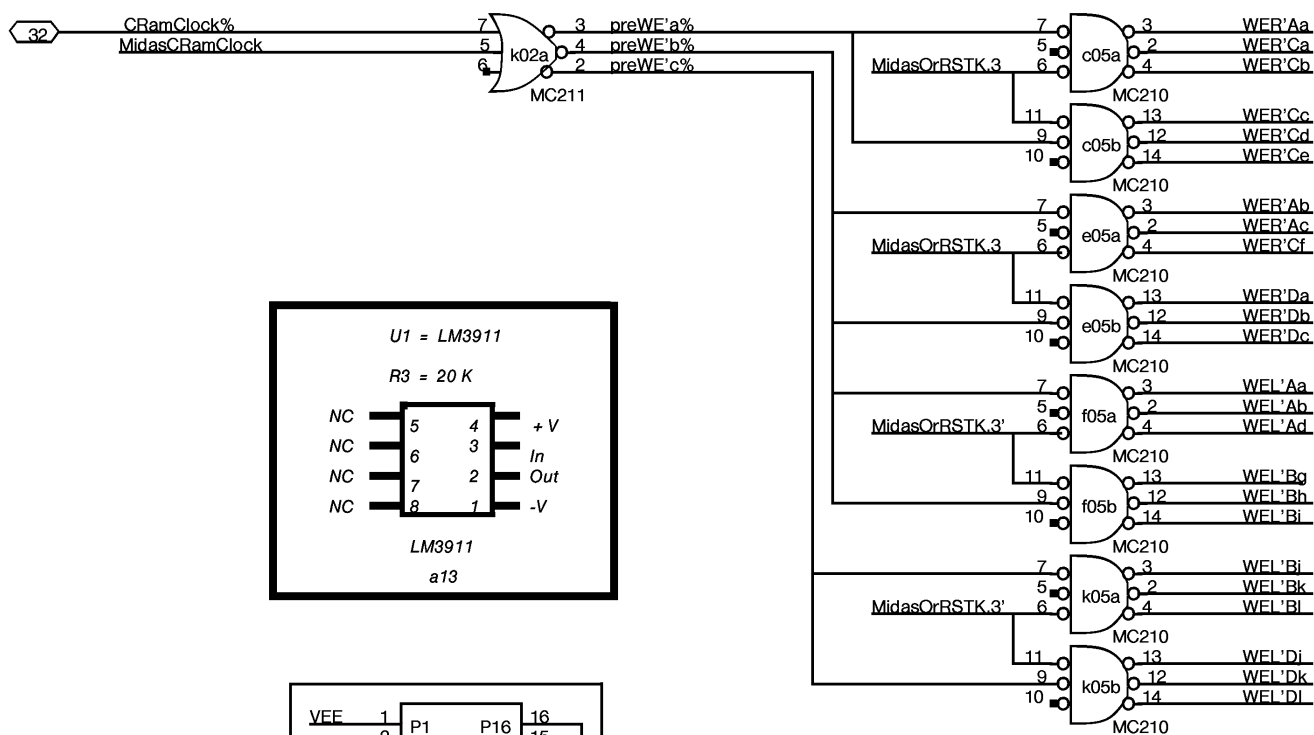
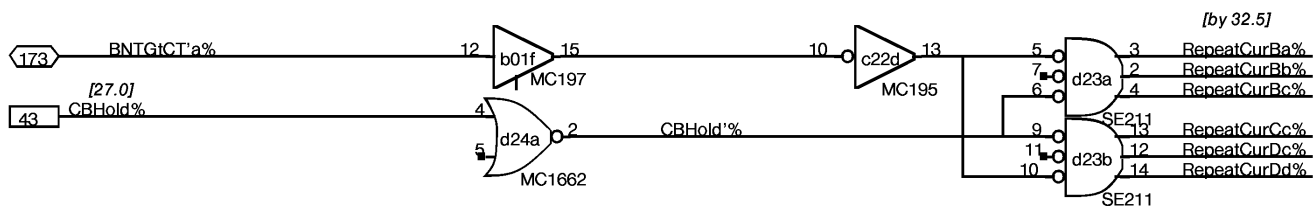
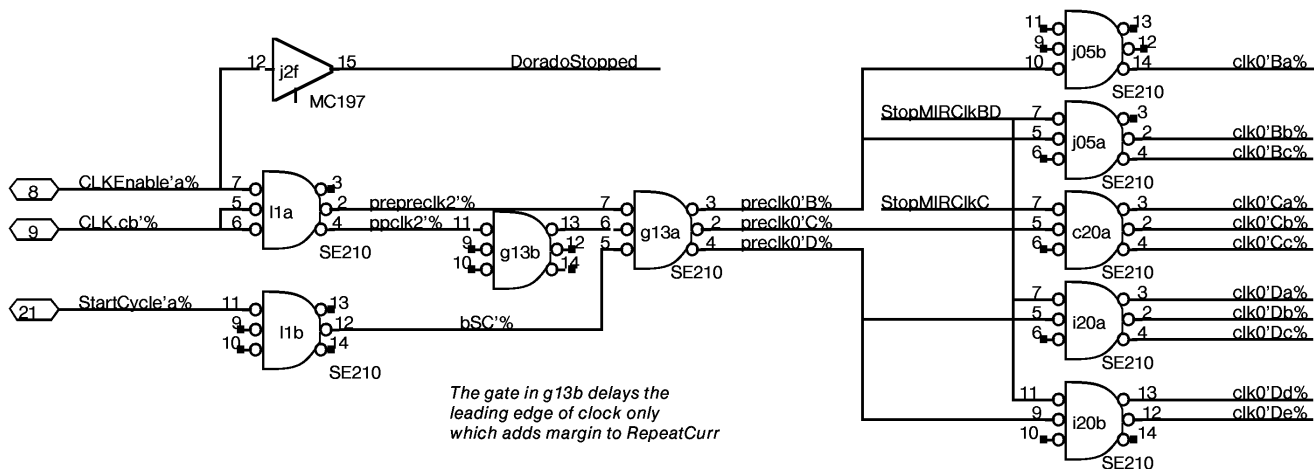


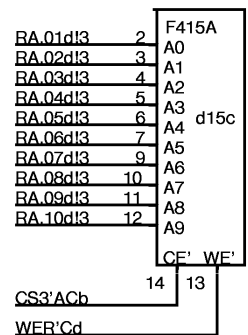
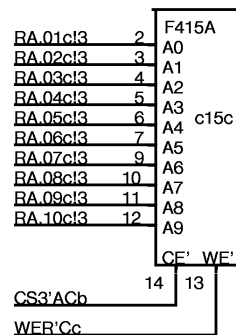
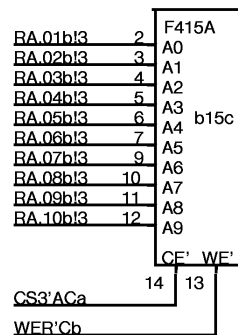
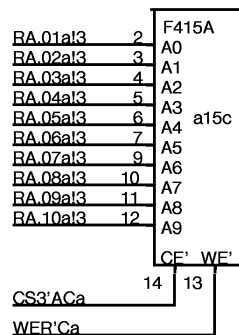
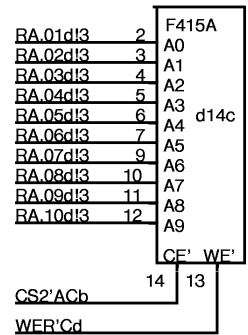
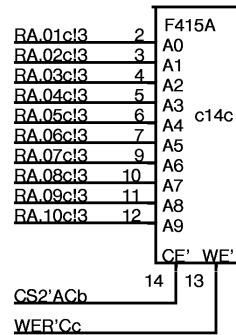
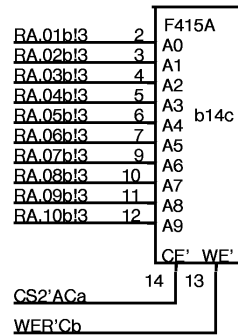
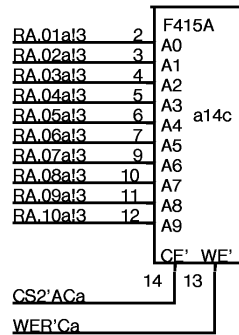
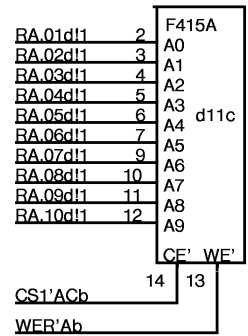
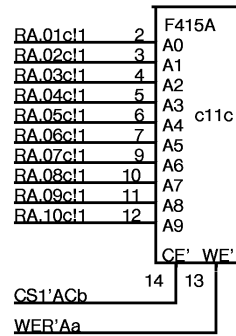
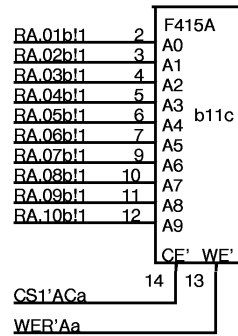
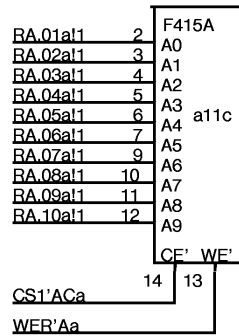
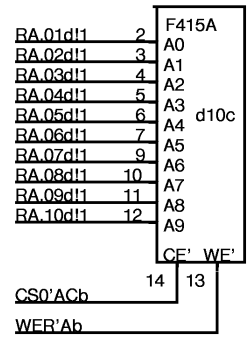
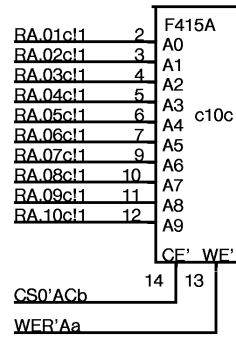
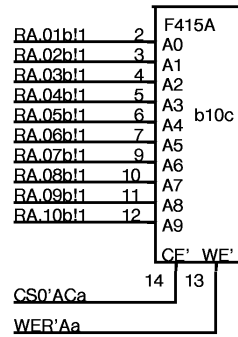
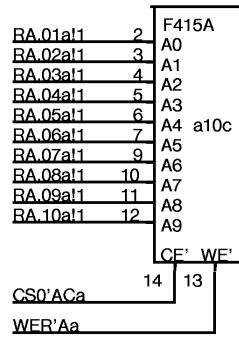
Figure 1 shows four schematic diagrams of 10-bit shift registers, labeled F415A, F415B, F415C, and F415D. Each register has 10 data inputs (A0-A9), a clock input (A0), and a control input (A9). The outputs are labeled CS0'ACa, WFL'Aa, CS0'ACb, and WFL'Ab for F415A; CS0'ACa, WFL'Ab, CS0'ACb, and WFL'Ad for F415B; CS0'ACb, WFL'Ab, CS0'ACd, and WFL'Ad for F415C; and CS0'ACb, WFL'Ad, CS0'ACd, and WFL'Ad for F415D. The registers are connected in a chain, with the output of one register serving as the input for the next.

Figure 1 shows four schematic diagrams of the F415A variants: a07c, b07c, c07c, and d07c. Each diagram consists of a box representing the protein structure, with residues numbered 1 to 12. The residues are labeled as RA.01aI0 to RA.10aI0 for a07c, RA.01bI0 to RA.10bI0 for b07c, RA.01cI0 to RA.10cI0 for c07c, and RA.01dI0 to RA.10dI0 for d07c. The mutation site is indicated by a red box around the residue number. The mutation site is located at the C-terminus of the protein, between residues 13 and 14. The mutation site is labeled as CF' WE' for a07c, CF' WE' for b07c, CF' WE' for c07c, and CF' WE' for d07c. The mutation site is also labeled as CS1'ACa for a07c, CS1'ACa for b07c, CS1'ACb for c07c, and CS1'ACb for d07c. The mutation site is also labeled as WEL'Aa for a07c, WEL'Ab for b07c, WEL'Ab for c07c, and WEL'Ad for d07c.

Figure 1: Schematic representation of the four F415A mutant strains. The figure is divided into four panels, each representing a different mutant strain: a08c, b08c, c08c, and d08c. Each panel shows a 12-bp DNA sequence (RA.01a10 to RA.10a10) and a 14-bp DNA sequence (CS2'ACa to WEL'Aa) for strain a08c, and a 12-bp DNA sequence (RA.01b10 to RA.10b10) and a 14-bp DNA sequence (CS2'ACa to WEL'Ab) for strain b08c, and a 12-bp DNA sequence (RA.01c10 to RA.10c10) and a 14-bp DNA sequence (CS2'ACb to WEL'Ab) for strain c08c, and a 12-bp DNA sequence (RA.01d10 to RA.10d10) and a 14-bp DNA sequence (CS2'ACb to WEL'Ad) for strain d08c. The F415A mutation is indicated by a red box around the 12-bp sequence.

Figure 1: Schematic representation of the four F415A mutant strains. The figure shows four panels, each representing a different mutant strain: a09c, b09c, c09c, and d09c. Each panel contains a 12x2 grid of boxes. The first column of boxes in each panel lists the names of the 12 genes (RA.01a1 to RA.10a1) and the second column lists the names of the 12 genes (RA.01b1 to RA.10b1). The labels a09c, b09c, c09c, and d09c are placed to the right of the first column of boxes. The labels CF' and WF' are placed below the first column of boxes. The labels CS3'ACa, WEL'Aa, CS3'ACb, and WEL'Ad are placed below the second column of boxes. The labels 14 and 13 are placed to the right of the second column of boxes.

IMLH

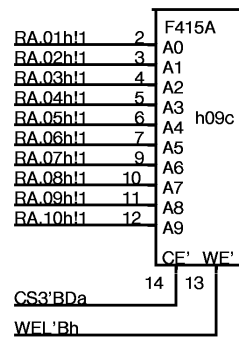
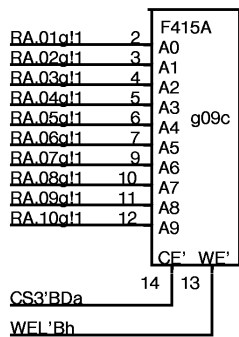
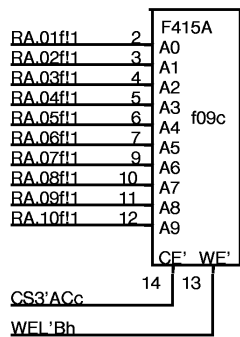
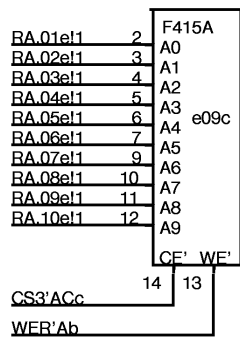
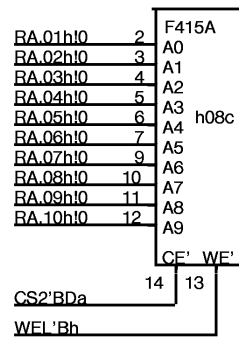
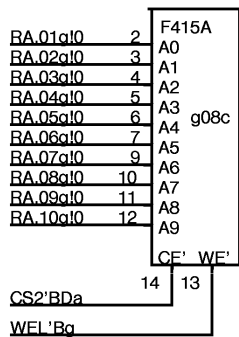
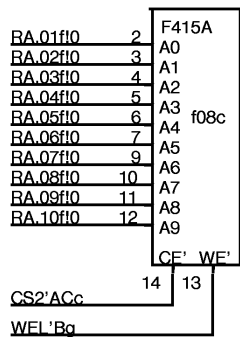
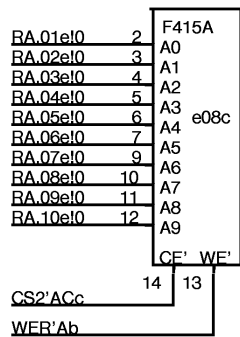
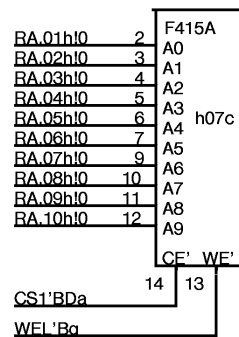
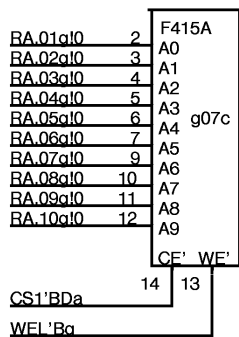
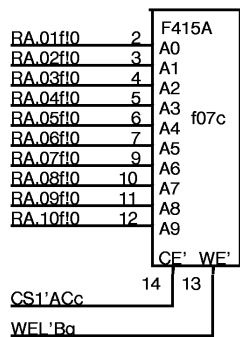
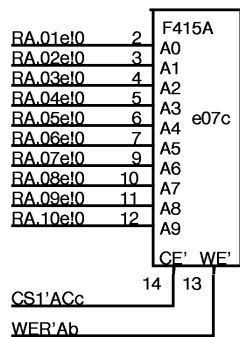
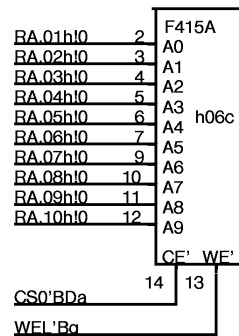
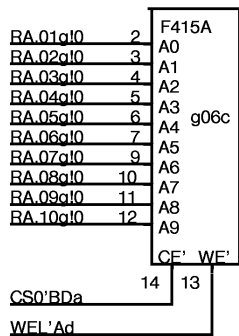
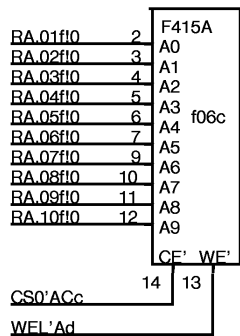
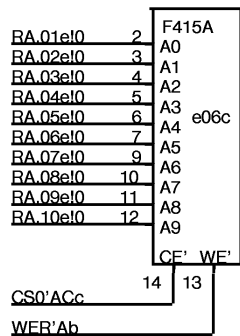


FF.0

FF.1

FF.2

FF.3

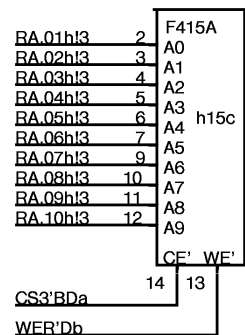
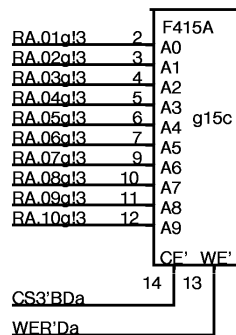
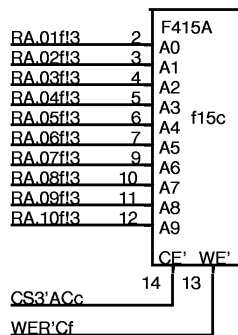
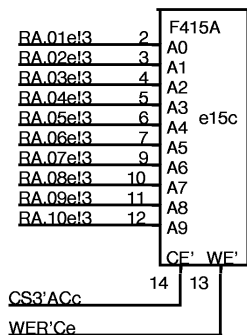
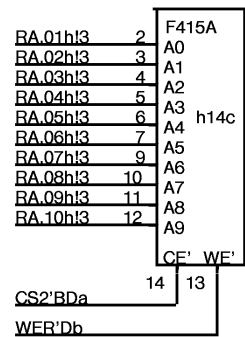
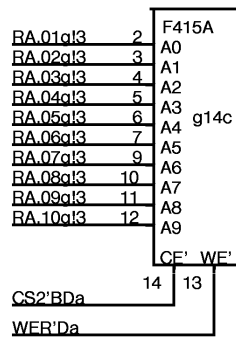
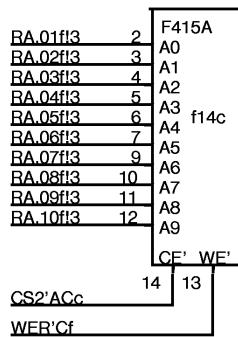
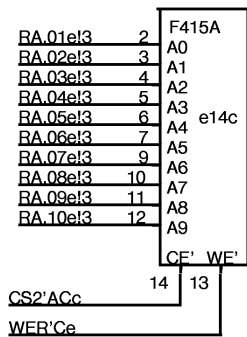
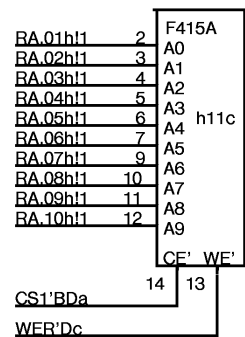
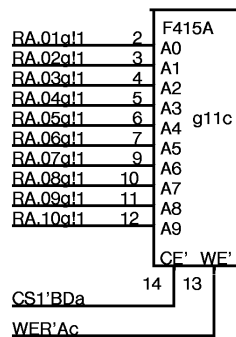
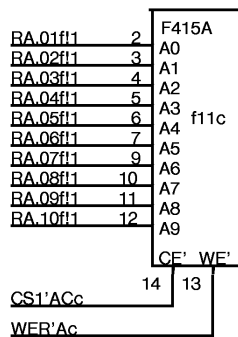
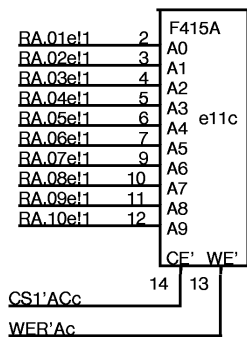
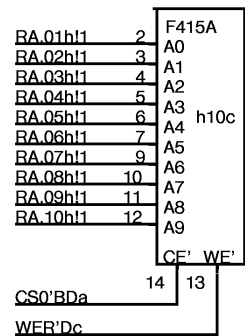
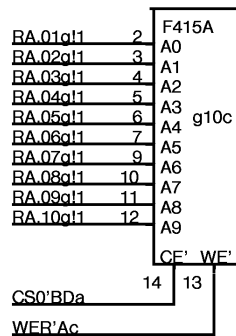
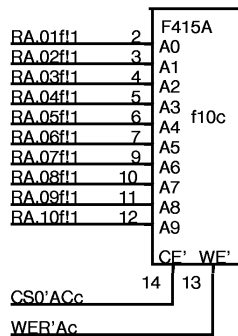
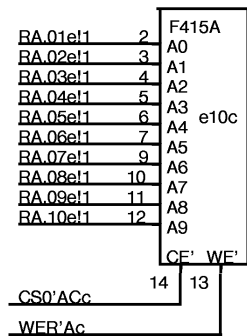


JCN.0

BSEL.0

BSEL.1

BSEL.2

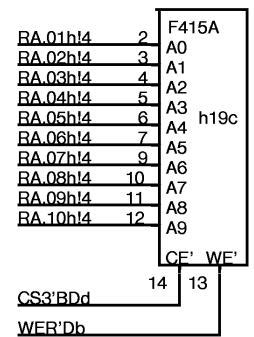
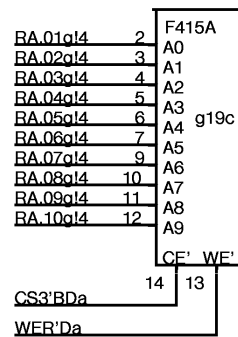
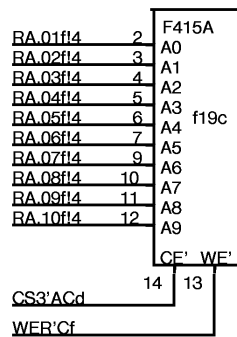
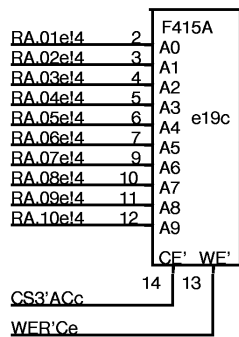
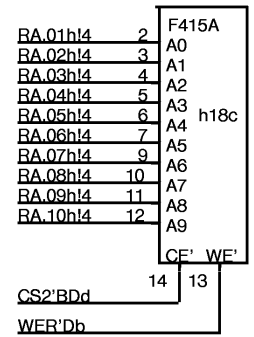
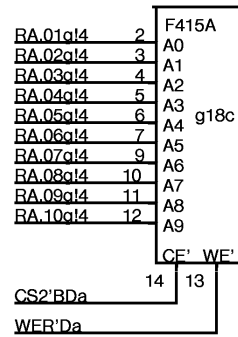
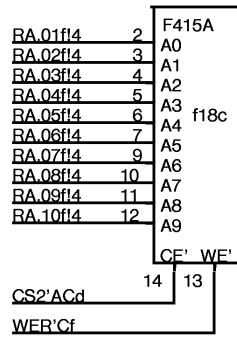
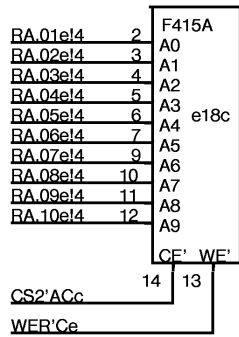
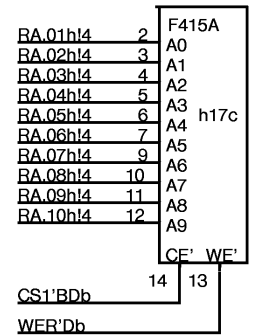
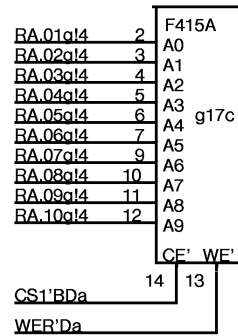
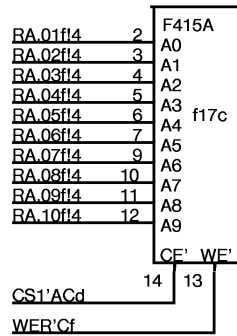
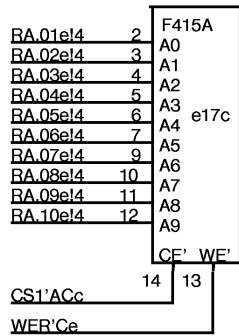
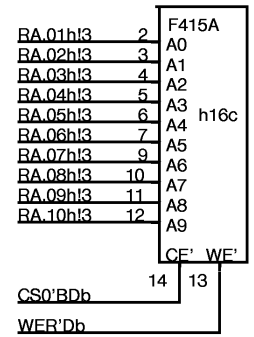
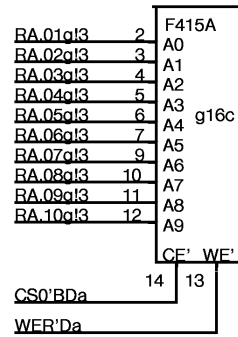
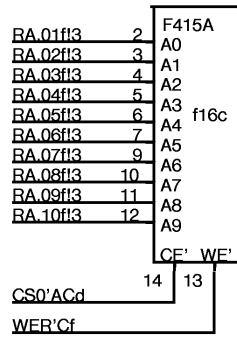
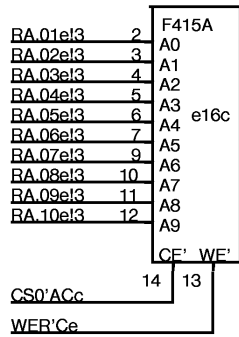


JCN.1

JCN.2

JCN.3

JCN.4

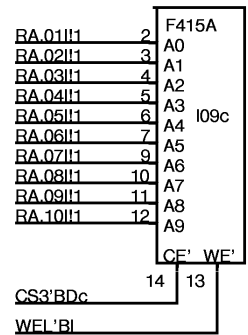
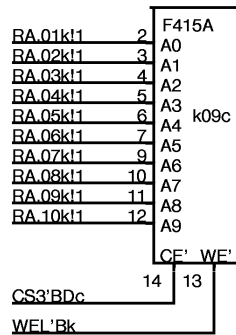
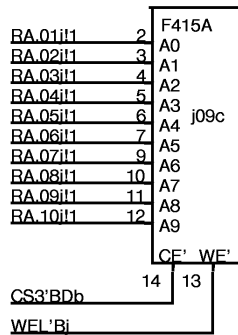
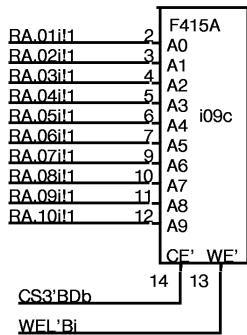
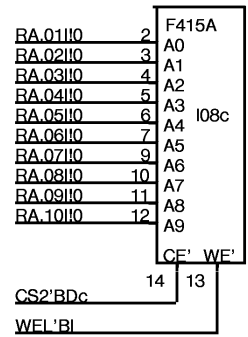
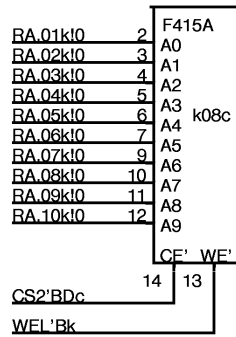
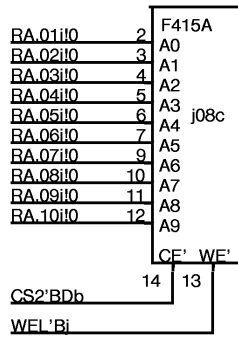
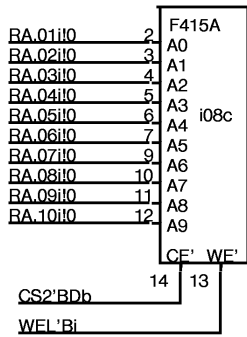
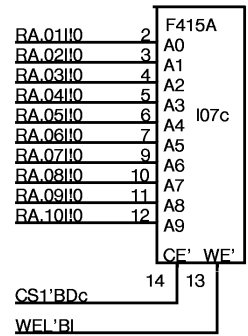
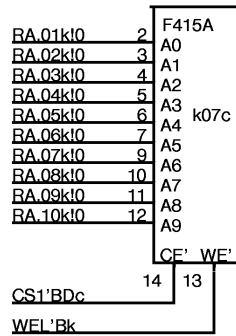
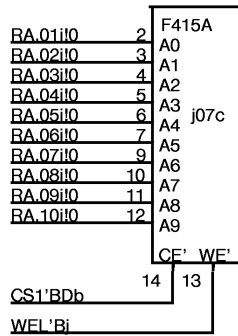
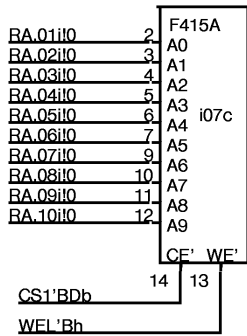
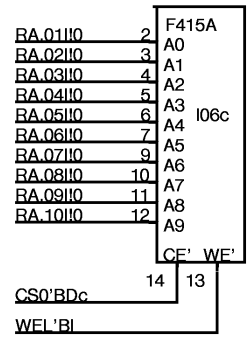
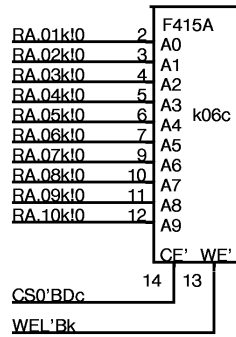
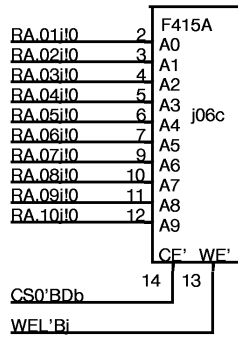
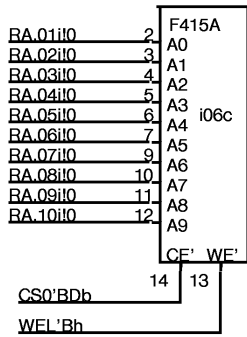


IMRH

BLK'

JCN.5

JCN.6

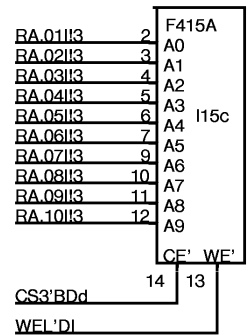
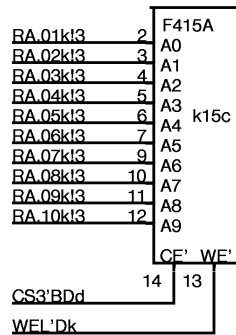
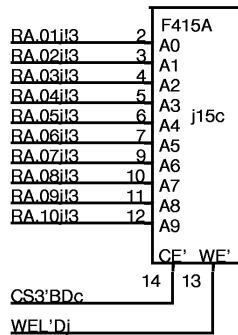
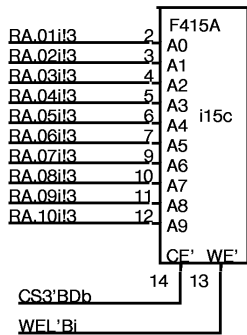
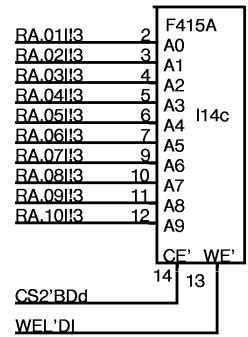
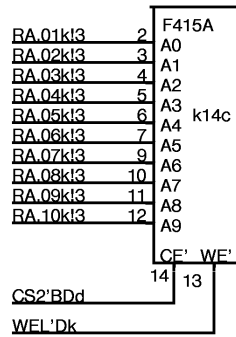
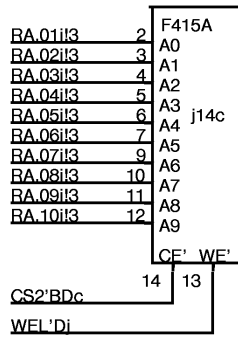
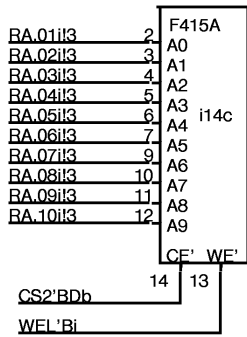
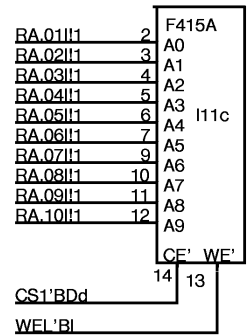
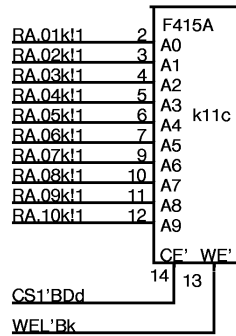
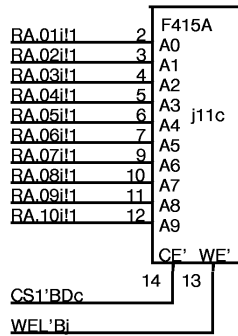
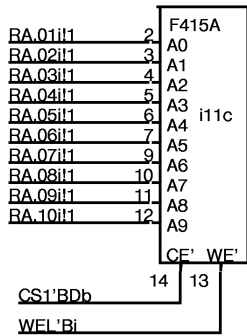
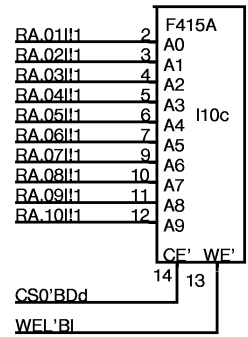
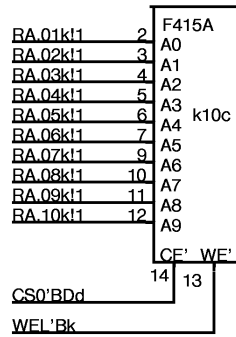
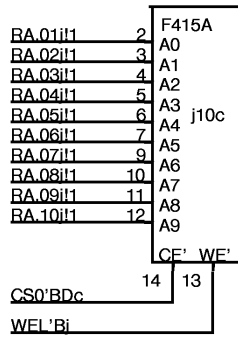
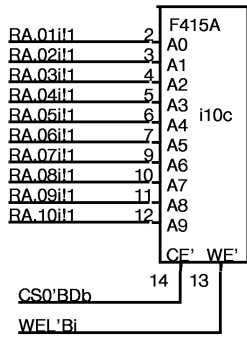


RSTK.0

RSTK.1

RSTK.2

RSTK.3



ALUF.0

ALUF.1

ALUF.2

ALUF.3

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6	F415A 0000 to 3777 Even		F415A 0	F415A 0	F415A 0	F415A 0			F415A 0	F415A 0	F415A 0	6		
7	ASEL 0000 to 3777 Odd		IMLH 1	JCN 1	JCN 1	BSEL 1			RSTK 1	RSTK 1	RSTK 1	7		
8	0 4000 to 7777 Even	1 4000 to 7777 Even	2 4000 to 7777 Even	0 4000 to 7777 Even	0 4000 to 7777 Even	1 4000 to 7777 Even	2 4000 to 7777 Even	2 4000 to 7777 Even	0 4000 to 7777 Even	1 4000 to 7777 Even	2 4000 to 7777 Even	3 4000 to 7777 Even	8	
9	0 4000 to 7777 Odd	1 4000 to 7777 Odd	2 4000 to 7777 Odd	0 4000 to 7777 Odd	0 4000 to 7777 Odd	1 4000 to 7777 Odd	2 4000 to 7777 Odd	2 4000 to 7777 Odd	0 4000 to 7777 Odd	1 4000 to 7777 Odd	2 4000 to 7777 Odd	3 4000 to 7777 Odd	9	
10	0	0	0	0	0	0	0	0	0	0	0	0	10	
11	FF 1			JCN 1	JCN 1	JCN 1	JCN 1		ALUF 1	ALUF 1	ALUF 1	ALUF 1	11	
12	RA 211 7	RA 211 7	RA 211 7	RA 211 7	RA 211 7	RA 211 7	RA 211 7	RA 211 7	RA 211 7	RA 211 7	RA 211 7	RA 211 7	12	
13	TEMP SENSE	RA 211 7	RA 211 7	RA 211 7	RA 211 7		preclk0 11,b 210	RA 211 7	RA 211 7	RA 211 7	RA 211 7	RA 211 7	Midas 107 12	13
14	0 2	1 2	2 2	3 2	1 2	2 2	3 2	4 2	0 2	1 2	2 2	3 2	14	
15	0 3	1 3	2 3	3 3	1 3	2 3	3 3	4 3	0 3	1 3	2 3	3 3	15	
16	F415A 0	F415A 0	F415A 0	F415A 0	F415A 0	F415A 0	F415A 0	F415A 0	F415A 0	F415A 0	F415A 0	F415A 0	16	
17	FF 1			IMRH 1	BLK' 1		JCN 1		JCN 1		LC 1	LC 1	17	
18	4 2	5 2	6 2	7 2	0 2	1 2	2 2	3 2	4 2	5 2	6 2	7 2	18	
19	0 3	1 3	2 3	3 3	0 3	1 3	2 3	3 3	4 3	5 3	6 3	7 3	19	
20		CLK0'Cx 11,b 210	MUbdxxx 164 12	dRA 04' & 05' 1662 6	dRA 06' & 07' 1662 6	dRA 08' & 09' 1662 6	dRA 10' & 11' 1662 6	CLK0'Dx 210 11	PRITY 170 9	INV 8,7,4,d,e,8 195	Midas 176 12		20	
21		MUbdxxx 164 12	CSEL 101 8	CSEL 101 8	dRA 00' & 01' 1662 6	dRA 02' & 03' 1662 6	CSEL 101 8	CSEL 101 8	PRITY 170 9	MidasSW 101 11			21	
22	dXXXBuf 3,3,3,4,4,4 197	INV 8,4,4,11, 195 12,8	CSEL 101 8	CSEL 101 8	MidasBNPC ManClk5 176 6	MidasBNPC ManClk6 176 6	CSEL 101 8	CSEL 101 8	dXXXBuf 1,1,2,2,e,f 197		Midas 176 12		22	
23	Local ASEL 3,c 231	dXXXBuf 4,4,4,4,4,5 197 4	RepeatCur 211 11	TNIA 164 12	MU 164 12	TNIA 164 12	BNPC 164 12	BNPC 164 12	dXXXBuf 2,2,3,3,2,2 197 2	MU MIR 164 12	MU MIR 164 12	Midas 164 12	23	
24	ASEL 0-1 231 3	ASEL 2 IMLH 231 3	b,c,d CbHold 1662 11	dXXXBuf 197 5	dXXXBuf 5,5,3,4,e,f 197 5	BSEL 0-1 231 2	BSEL2,LC0 231 2	LC 1-2 231 3	ALUF 0-1 231 1	ALUF 2-3 231 2		12,b,12 ,12 103	24	
C	a 11	b 26	c 39	d 55	e 70	f 86	99 g	114 h	129 i	143 i	159 k	174 l	D	